Muscle Activation During Selected Strength Exercises in Women With Chronic Neck Muscle Pain

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Background and Purpose. Muscle-specific strength training has previously been shown to be effective in the rehabilitation of chronic neck muscle pain in women. The aim of this study was to determine the level of activation of the neck and shoulder muscles using surface electromyography (EMG) during selected strengthening exercises in women undergoing rehabilitation for chronic neck muscle pain (defined as a clinical diagnosis of trapezius myalgia).

Subjects. The subjects were 12 female workers (age=30–60 years) with a clinical diagnosis of trapezius myalgia and a mean baseline pain intensity of 5.6 (range=3–8) on a scale of 0 to 9.

Method. Electromyographic activity in the trapezius and deltoid muscles was measured during the exercises (lateral raises, upright rows, shrugs, one-arm rows, and reverse flys) and normalized to EMG activity recorded during a maximal voluntary static contraction (MVC).

Results. For most exercises, the level of muscle activation was relatively high (>60% of MVC), highlighting the effectiveness and specificity of the respective exercises. For the trapezius muscle, the highest level of muscle activation was found during the shrug (102±11% of MVC), lateral raise (97±6% of MVC), and upright row (85±5% of MVC) exercises, but the latter 2 exercises required smaller training loads (3–10 kg) compared with the shrug exercise (20–30 kg).

Discussion and Conclusion. The lateral raise and upright row may be suitable alternatives to shrugs during rehabilitation of chronic neck muscle pain. Several of the strength exercises had high activation of neck and shoulder muscles in women with chronic neck pain. These exercises can be used equally in the attempt to achieve a beneficial treatment effect on chronic neck muscle pain.

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Musculoskeletal pain is the most common condition treated by physical therapists and contributes substantially to health care costs and sick leave. The occurrence of neck muscle pain has been increasing in recent decades.[1] Neck pain originating from the upper trapezius muscle (trapezius myalgia) is common in female office workers and occurs in other occupations that involve repetitive and monotonous work tasks.[2–4] Sustained activity of low-threshold motor units of the trapezius muscle day after day leads to disturbances in the intramuscular biochemical milieu, and myalgia often develops.[5–7]

Interventions that aim to relieve muscle pain often involve physical activity,[8] and various forms of physical training are known to relieve neck muscle pain to a certain extent.[9–14] Recently, high-intensity strength (force-generating capacity) training specifically for the neck and shoulder muscles for as little as 20 minutes 3 times a week was shown to markedly reduce pain in women with trapezius myalgia.[15] The results of that study along with a recent review[16] suggest that a high level of activation of the painful muscles (ie, a high training intensity) plays an important role in the rehabilitation process. Based on functional anatomy of the neck/shoulder complex, it is likely that isolated shoulder elevation ("shrugs") is the most specific exercise for the upper trapezius muscle,[17] a conclusion that also is supported by electromyographic (EMG) measurements.[18,19]

Shrug exercises, therefore, frequently are recommended to effectively target the trapezius muscle during rehabilitation.[18,19] However, the small external moment arm results in training weights that are 4 to 5 times larger than those of other types of neck or shoulder exercises (eg, shoulder abduction exercises). This is a major practical problem during rehabilitation because handgrip and low back strength may become limiting factors in handling these heavy weights in shrug exercises. Furthermore, for home-based rehabilitation, it may be more convenient to have dumbbells in the range of 3 to 8 kg for exercises such as the lateral raise, reverse fly, and upright row compared with 20 to 30 kg for the shrug exercise. Therefore, it is relevant to investigate whether other strengthening exercises with lower absolute weights but longer external moment arms can induce levels of muscle activation in the trapezius muscle that are as high as those of the shrug exercise.

The aim of this study was to determine the level of muscle activation of the neck and shoulder muscles with surface EMG during selected strengthening exercises in women undergoing rehabilitation for trapezius myalgia. This was done by comparing responses in 5 different types of exercise, all of which are thought to be effective in the activation of neck/shoulder muscles: (1) shrugs, (2) one-arm rows, (3) upright rows, (4) reverse flys, and (5) lateral raises.

Materials and Method

Subjects

This study was part of a large intervention study performed in Copenhagen, Denmark, during the period from September 2005 to March 2006.[15] Briefly, a questionnaire was sent to 802 female workers between the ages 30 and 60 years from 7 workplaces. The work tasks at these companies were monotonous and repetitive, and 82% of the employees worked at the computer for more than three fourths of their working time. Of these 802 workers, 306 agreed to participate in the study, 214 declined, and 282 did not reply. Out of the 306 positive replies, 48% qualified as neck or shoulder cases based on the questionnaire replies. The inclusion criteria were: (1) pain or discomfort in the neck/shoulder region for more than 30 days during the past year but no more than 3 bodily regions of pain or discomfort in order to exclude generalized musculoskeletal diseases; (2) the subject's pain or discomfort should be rated at least "quite a lot" on a 5-point ordinal scale ("a little," "somewhat," "quite a lot," "much," and "very much"); (3) the pain or discomfort should be frequent (at least once a week on an ordinal scale of "seldom," "once a week," "2–3 times per week," "almost all the time"); and (4) the intensity of the pain or discomfort should be rated at least 2 on a scale from 0 to 9, where 0 is "no pain" and 9 is "the worst imaginable pain."[20–22]

A clinical diagnosis confirmed that 69% of the questionnaire-defined neck/shoulder cases had trapezius myalgia. The main clinical criteria for trapezius myalgia were: (1) pain in the neck area, (2) tightness of the upper trapezius muscle, and (3) palpable tender spots in the upper trapezius muscle.[2,23] Women with serious conditions such as previous trauma, life-threatening diseases, fibromyalgia, whiplash, or arthritis in the neck and shoulder were excluded. The included subjects—all with a clinical diagnosis of trapezius myalgia—were randomly assigned to 1 of 3 intervention groups, of which one group performed specific strength training with the exercises described below for 20 minutes 3 times a week with relative loadings of their 8–12 repetition maximum (RM).

In the present study, EMG measurements (mean±SD) were performed on 12 of the women in the strength training group (mean age=43±9 years, mean weight=73±13 kg, mean height=166±7 cm). Based on the screening questionnaire replies at baseline, 33% of subjects experi-

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enced pain or discomfort in the neck/shoulder region for 31 to 90 days, 50% of the subjects experienced pain or discomfort for more than 90 days, and 17% of the subjects experienced pain or discomfort every day. The pain or discomfort was "quite a lot" in 42% of the subjects, "much" in 50% of the subjects, and "very much" in 8% of the subjects. The frequency of pain or discomfort was "2–3 times per week" in 58% of the subjects, and "almost all the time" in 42% of the subjects. The mean (±SD) pain intensity in the neck/shoulder region was 5.6±1.5 on a scale of 0 to 9 and ranged between 3 to 8. The women had been undergoing rehabilitation with the exercises described below for 8 to 10 weeks at the time of testing, which ensured that all subjects were highly familiar with the technique and proper training load of each specific exercise. All subjects were informed about the purpose and content of the project and gave written informed consent to participate in the study that conformed to the Declaration of Helsinki and was approved by the local ethical committee (KF 01-138/04).

Figure 1.
Illustration of the neck or shoulder rehabilitation exercises: (A) shrugs (SHR), (B) one-arm rows (ONE), (C) upright rows (UPR), (D) reverse flys (REV), and (E) lateral raises (LAT).

Exercise Description

The aim of the exercise program was to target the painful trapezius muscle with simple and inexpensive training equipment. Electromyographic activity during 5 different strengthening exercises was recorded (Fig. 1). All exercises were performed dynamically in a controlled manner: lifting (~1.5 seconds) and lowering (~1.5 seconds) the dumbbell without sudden jerks or accelerations. Visual inspection ensured that improper sets were immediately terminated and an extra set was performed. For all exercises tested, the heaviest weight that could be lifted for an 8-repetition maximum (8-RM) was used, which was determined 1 week prior to testing. Two sets of 3 repetitions (ie, a total of 6 repetitions) of each exercise were performed in randomized order by drawing a piece of paper from a box, with rest periods of 2 minutes between sets to avoid muscle fatigue. One set of each exercise was completed before the second set of each exercise was performed.

1. *Shrugs*. The subject stood erect and held the dumbbells to the side, and then elevated the shoulders while focusing on contracting the upper trapezius muscle (Fig. 1A).

2. *One-arm rows*. The subject bent her torso forward to approximately 30 degrees from horizontal with one knee on the bench and the other foot on the floor. The subject then pulled the dumbbell toward the ipsilateral lower rib, while the contralateral arm was maintained in extension and supported the body on the bench (Fig. 1B).

3. *Upright rows*. The subject stood erect and held the dumbbells while the arms hung relaxed in front of the body. The dumbbells were lifted toward the chest in a vertical line close to the body while the elbows were flexed and the shoulder abducted. The elbows pointed out and upward (Fig. 1C).

4. *Reverse flys*. The subject lay on her chest at a 45-degree angle from horizontal with the arms pointing toward the floor. The dumbbells were raised until the upper arms were horizontal, while the elbows were in a static, slightly flexed position (~5°) during the entire range of motion (Fig. 1D).

5. *Lateral raises*. The subject stood erect and held the dumbbells to the side, and then abducted the shoulder joints until the upper arms were horizontal. The elbows

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were in a static, slightly flexed position (~5°) during the entire range of motion (Fig. 1E).

Maximal Voluntary Static Contraction

Maximal voluntary static contractions (MVCs) of the shoulder were performed according to standardized procedures during static elevation and extension, abduction, and flexion to determine the maximal EMG amplitude for the upper trapezius muscles and the posterior, medial, and anterior parts of the deltoid muscle, respectively.[24] Three MVCs were performed for each muscle, and the trial with the highest EMG amplitude was used for normalization of the peak EMG in the rehabilitation exercises.

EMG Signal Sampling

Electromyographic signals were recorded from the upper trapezius muscle and the anterior, medial, and posterior parts of the deltoid muscle with a bipolar surface EMG configuration (Neuroline 720 01-K*) and an interelectrode distance of 2 cm. Before affixing the electrodes, the skin of the respective area was prepared with a scrubbing gel (Acqua gel†) to effectively lower the impedance to less than 10 kΩ. For the trapezius muscle, the electrodes were positioned 2 cm medially from the midpoint between the acromion and the seventh cervical vertebra. The 3 separate parts of the deltoid muscle were located by palpation, and the electrodes were placed at one fourth the distance from the acromion to the olecranon.[25] The EMG electrodes were connected directly to small preamplifiers located near the recording site. The raw EMG signals were led through shielded wires to instrumental differentiation amplifiers (with a bandwidth of 10 to 500 Hz and a common mode rejection ratio better than 100 dB), sampled at 1,000 Hz using a 16-bit A/D-converter (DAQCard-Al-16XE-50‡), and stored on a laptop for further analysis.

EMG Analysis

During later analysis, all raw EMG signals obtained during MVCs as well as during the dynamic rehabilitation exercises were digitally filtered using linear EMG envelopes, which consisted of: (1) high-pass filtering at 10 Hz, (2) full-wave rectification, and (3) low-pass filtering at 10 Hz. The filtering algorithm was based on a fourth-order, zero phase lag Butterworth filter.[26] For each contraction of the dynamic exercises as well as the MVCs, the highest value of integrated EMG over any 500-millisecond interval was used as peak EMG. During the training exercises, peak EMG for each of the 2 × 3 repetitions was determined, and the average value of these 6 repetitions was then normalized to the peak EMG obtained during the MVC of each respective muscle.

Data Analysis

For each muscle, a one-way analysis of variance with a Tukey corrected *post hoc* test was used to determine whether differences existed in the level of muscle activation of the 5 different exercises. The intra-day reliability in peak EMG amplitude was determined by calculating the intraclass correlation coefficient (ICC [3,1]), between the 2 sets of each exercise. The level of significance was set at .05, and all values were reported as means ± SE. Statistical analysis was conducted with SAS version 9.§

Results

Training Load

The 8-RM loads for the rehabilitation exercises were: shrugs=24±1.3 kg, one-arm rows=15±0.8 kg, upright rows=8.6±0.6 kg, lateral raises=5.2±0.3 kg, and reverse flys=4.8±0.4 kg.

Trapezius Muscle Activation

Numbers in parentheses express EMG activity during the respective exercises normalized to measurements of EMG activity obtained during the MVC, which was 971±88 μV for the upper trapezius muscle. The level of trapezius muscle activation was statistically higher during shrugs (102±11%) and lateral raises (97±6%) compared with reverse flys (72±4%) and one-arm rows (41±6%) ($P<.01$); furthermore, the level of trapezius muscle activation during upright rows (85±5%) was statistically higher compared with one-arm rows (41±6%) ($P<.01$) (Fig. 2A). Although the level of activation during the shrugs and lateral raises was numerically higher than during the upright rows, it was not statistically significant ($P=.14$). Figure 3 shows a representative EMG recording from one of the subjects for the lateral raise and shrug exercises.

Deltoid Muscle Activation

Numbers in parentheses express EMG activity during the respective exercises normalized to the measurements of EMG activity obtained during the MVC, which was 450±59, 492±56, and 649±84 μV for the posterior, medial, and anterior part of the deltoid muscle, respectively. The level of muscle activation in the posterior part of the deltoid muscle was significantly higher during reverse flys (102±9%) compared with shrugs (71±5%), upright rows (69±8%), and lateral raises (59±7%) ($P<.01$); furthermore, the level of posterior deltoid muscle activation was higher during one-arm rows

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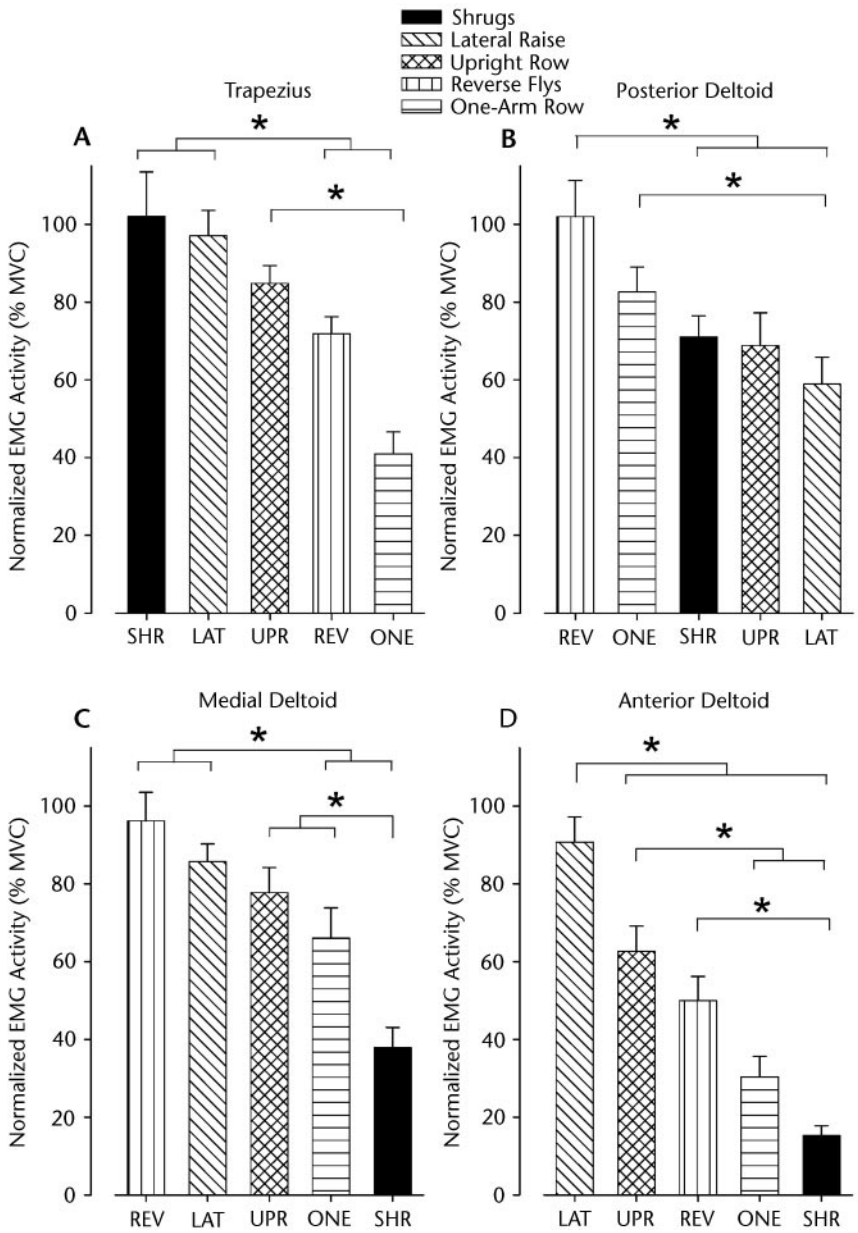

Figure 2.
The highest level of muscle activation (normalized electromyographic [EMG] activity) was found in: (A) the trapezius muscle during shrug (SHR) and lateral raise (LAT) exercises, (B) the posterior deltoid muscle during the reverse fly (REV) exercise, (C) the medial deltoid muscle during the REV and LAT exercise, and (D) the anterior deltoid muscle during the LAT exercise. Asterisk (*) indicates significant difference in normalized EMG between exercises ($P<.01$). ONE=one-arm row exercise, UPR=upright row exercise, MVC=maximal voluntary static contraction.

(83±6%) compared with lateral raises (59±7%) ($P<.01$) (Fig. 2B).

In the medial part of the deltoid muscle, the level of muscle activation was higher during reverse flys (96±7%) and lateral raises (86±5%) compared with one-arm rows (66±8%) and shrugs (38±5%) ($P<.01$); furthermore, the level of medial deltoid muscle activation was higher during upright rows (78±6%) and one-arm rows (66±8%) compared with shrugs (38±5%) ($P<.01$) (Fig. 2C).

In the anterior part of the deltoid muscle, the level of muscle activation was higher during lateral raises (91±6%) compared with the other 4 exercises ($P<.01$) and was higher during upright rows (63±6%) compared with one-arm rows (30±5%) and shrugs (15±2%) ($P<.01$). Furthermore, the level of anterior deltoid muscle activation was higher during reverse flys (50±6%) compared with shrugs (15±2%) ($P<.01$) (Fig. 2D).

Intra-day Reliability

For each exercise and muscle, the peak EMG amplitude of the 2 sets performed was highly reliable (ICC [3,1]=.94–1.00, $P<.001$) (Table).

Discussion

In this study, relatively high levels of muscle activation in the trapezius and deltoid muscles were found for several of the exercises used for treatment of trapezius myalgia. These results document the effectiveness of the investigated exercises and suggest that there are several ways to reach a high level of muscle activation in women with myalgia.

It is generally agreed that strength training intensities of at least 60% should be used for effective muscular adaptations to occur and that higher intensities yield proportionally greater adaptations.[27] Strength training intensity is defined as a percentage of the strength of a maximal voluntary muscle contraction,[27] and can be roughly estimated as a percentage of the maximal EMG amplitude during an MVC.[28] For the trapezius muscle, a linear force-EMG relationship has been documented[29]; therefore, any difference in normalized EMG amplitude between exer-

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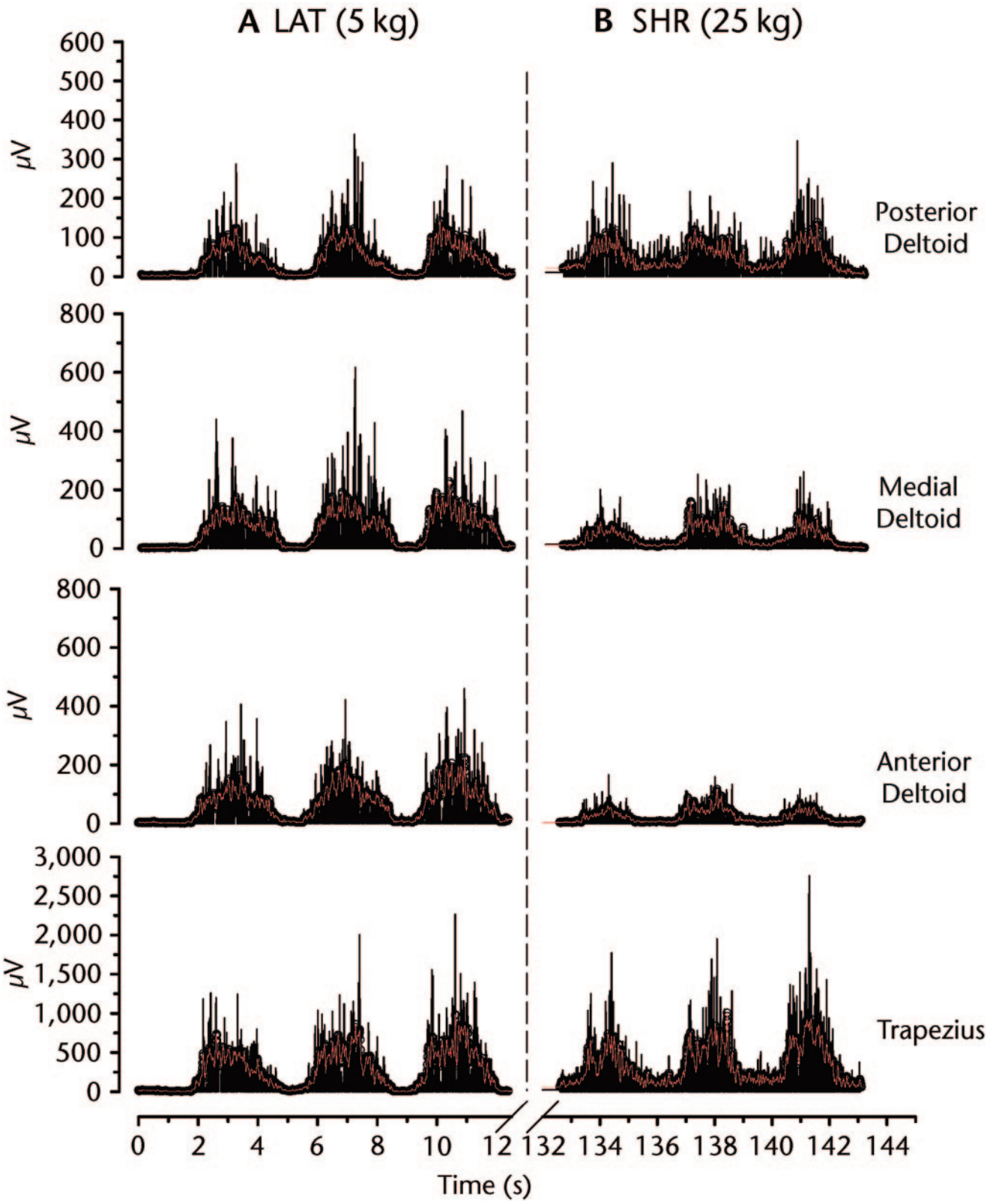

Figure 3.
A representative raw rectified electromyographic recording from one of the subjects during (A) a lateral raise (LAT) exercise with 5 kg and (B) a shrug (SHR) exercise with 25 kg. The filtered signal is overlaid (red tracing) on the raw recording.

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cises reflects relative differences in levels of muscle force. It is positive to note that 4 out of 5 exercises induced trapezius EMG amplitudes that were above 60% of MVC (Fig. 2A), which means that a broad range of specific strengthening exercises can be used for rehabilitation of trapezius myalgia.

A major finding of the present study was that the lateral raise exercise induced similarly high levels of trapezius muscle activation compared with the shrug exercise (Fig. 2A), despite the fact that the average training weight during the lateral raise exercise was only 5 kg compared with 24 kg during the shrug exercise. This finding has important practical relevance for rehabilitation of trapezius myalgia, because grip strength and low back strength may become limiting factors during heavy-load shrugs, especially for those with symptoms that also appear in the low back and hip or knee. During the lateral raise exercise, the upper trapezius muscle rotates the scapula upward,[30] and, together with a long external moment arm, this may explain the high level of trapezius muscle activation in spite of relatively low external loadings. Furthermore, the upright row exercise induced levels of trapezius muscle activation that were not significantly lower than the levels of the shrug and lateral raise exercises. Thus, lateral raise and upright row exercises may be offered as alternatives to shrugs for patients with limited grip strength and low back and hip or knee symptoms. It is important to note that shoulder abduction exercises (eg, lateral raises and upright rows) can be demanding on the rotator cuff muscles and can be associated with impingement-like symptoms due to compression of the supraspinatus tendon. In these cases, shrugs may be appropriate for patients who have problems abducting the arm (eg, due to shoulder joint impingement). Although not specifically investigated in the present study, a combination of the exercises may be preferable when possible. Based on the present findings, we concluded that several of the exercises used will lead to a marked and relevant increase in activation of muscles with chronic pain and, therefore, contribute to a positive treatment effect.

Table.
Intra-day Reliability in Peak Electromyography Amplitude, Expressed as the Intraclass Correlation Coefficient (ICC [3,1]) Between the First and Second Set of Each Exercise

Exercise	Muscle			
	Anterior Deltoid	Medial Deltoid	Posterior Deltoid	Trapezius
Lateral raise	.99	.98	.94	.97
Reverse fly	.99	1.00	.95	.99
One-arm raise	.96	1.00	.98	.97
Shrug	.95	.99	.98	.99
Upright row	.98	.98	1.00	.99

In contrast to the present findings, Ekstrom and coworkers[18] found that shrugs induced higher levels of trapezius muscle activation compared with shoulder abduction exercises. The subjects in that study, however, were healthy and untrained. In contrast, the women who participated in the present study were clinically diagnosed with trapezius myalgia and were part of an ongoing rehabilitation program. Learning and coordination of a specific exercise have marked influence on the level of muscle activation during the initial weeks of training, especially when more complex exercises are used.[31] Therefore, training should be performed for at least a few weeks before measuring the level of muscle activation to be able to compare the long-term benefits of different exercises.

In the present study, the women had been undergoing rehabilitation with the exercises for 8 to 10 weeks at the time of testing, which ensured that all subjects were highly familiar with the technique and proper training load of each specific exercise. Intra-day reliability of EMG amplitude during the exercises was very high (ICC [3,1]=.94–1.00), which indicates that the subjects were highly familiar with each specific exercise. In comparison, lower ICCs (.30–.94) for the upper trapezius muscle were reported in a previous study with untrained subjects.[18]

The deltoid muscle plays a major role as a prime mover of the glenohumeral joint, and some patients with neck or shoulder conditions may need rehabilitation of specific parts of the deltoid muscle. More importantly, perhaps, some patients may need to avoid activation of certain parts of the shoulder muscles (eg, in severe shoulder disorders such as shoulder joint impingement or rotator cuff tendinitis). For the posterior deltoid muscle, the highest level of activation was achieved by using the exercises that involved a forward incline of the body (ie, reverse flys and one-arm rows). The lateral raise exercise is traditionally considered the most specific exercise for the medial deltoid muscle. However, the reverse fly exercise induced similarly high levels of muscle activation in this part of the deltoid muscle. The lateral raise exercise appeared to be more specific compared with the other exercises with regard to the anterior deltoid muscle, probably

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due to outward rotation of the humerus during abduction of the arm in this exercise.

A limitation of the present study is that EMG measurements were performed only after 8 to 10 weeks of training. Therefore, it is not known whether similarly high levels of muscle activation were achieved during the initial weeks of training. Furthermore, to quantify the level of muscle activation, EMG activity during the dynamic exercises was normalized to measurements of EMG activity during the static MVCs. Given the inherent methodological limitations associated with surface EMG, only a rough estimate of the level of muscle activation can be inferred using this method.[32]

Conclusion

The lateral raise and upright row exercises can be used as alternatives to the shrug exercise during rehabilitation of chronic neck muscle pain (trapezius myalgia) because they result in relevant and marked increases in trapezius muscle activation compared with the shrug exercise. These 2 alternative exercises require less training weight (3–10 kg) compared with the shrug exercise (20–30 kg) and, from a practical standpoint, may be easier to carry out as regular training exercises. Finally, it should be recognized that these exercises are to be prescribed for conditions of myalgia (ie, pain from the muscles) and that the present study did not investigate the effect upon other, more severe neck or shoulder disorders.

Lars Andersen, Dr Kjær, and Christoffer Andersen provided concept/idea/research design. Lars Andersen, Dr Kjær, and Dr Zebis provided writing. Lars Andersen, Christoffer Andersen, Peter Hansen, and Klaus Hansen provided data collection. Lars Andersen, Christoffer Andersen, Dr Zebis, and Klaus Hansen provided data analysis. Lars Andersen, Peter Hansen, Klaus Hansen, and Dr Sjøgaard provided project management. Dr Sjøgaard provided fund procurement, facilities/equipment, and institutional liaisons. Christoffer Andersen, Peter Hansen, and Dr Sjøgaard provided subjects. Dr Kjær, Dr Zebis, and Dr Sjøgaard provided consultation (including review of the manuscript before submission).

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